Purdue SCALE – Laboratory Activity Form - Instructor

Basic Information
Course / Module: ADMF 122 – Industrial Electrical II

Lab Title:

Lab Number:

Instructor: ______________________________________

Date Assigned: ______________ Date Submitted: ______________

Student Name: ____________________________________

Team Members: ____________________________________

SCALE Competency Area(s)
Trusted Microelectronics

Embedded Systems

Hardware Assurance

Arduino Programming (Simplified C++)

Lab Objectives
1. Understand breadboarding basics

2. Analyze simple schematic diagrams

3. Program Microprocessors

4. Assemble digital circuits

5. Test motor control circuits

Required Equipment & Resources
• Elegco Arduino Super Starter Kit

• Computer with Arduino programming software

Software / Tools – Arduino IDE software (free download)- digital multimeter
References: https://www.arduino.cc/en/software/

https://us.elegoo.com/products/elegoo-uno-r3-super-starter-kit?srsltid=AfmBOorMWInXz4ROr4TSrhNVQ2_RJ3dKgmwaA67tV0lu0tKNIkOkfSU

Pre-Lab Requirements
Review assigned readings
Watch pre-lab video
Complete safety briefing

Procedure
Steps:

1. Identify components
2. Install power supply
3. Connect circuit components
4. Download program to Arduino
5. Test for proper operation

Observations: Record sequence of events and other important information

Data Collection & Results

Record voltages at different points in the sequence.
(example)Voltage when running clockwise, counter clockwise, full speed.

Analysis & Discussion

Modify the program as directed

Conclusion
1

2.

Instructor Verification
Lab Completed Successfully: ☐ Yes ☐ No

Instructor Signature: ______________________________

Date: ______________________